

Lorentz Transformation

$$\begin{cases} x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \\ y' = y \\ z' = z \\ ct' = \frac{ct - \frac{v}{c}x}{\sqrt{1 - v^2/c^2}} \end{cases}$$

K (track) stationary K' (train) speed +v
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Matrix form:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}$$

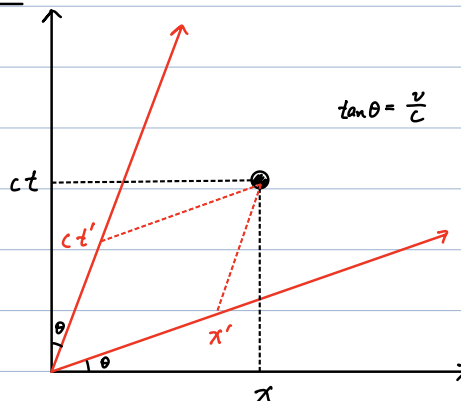
where

$\beta \equiv v/c$ $\gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}$
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Note: Change of frame $\leftrightarrow$ change of basis

$$\{\hat{e}_t, \hat{e}_x\} = \{\hat{e}_{t'}, \hat{e}_{x'}\} \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix}$$

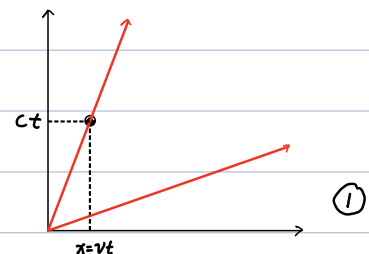
Minkowski diagram



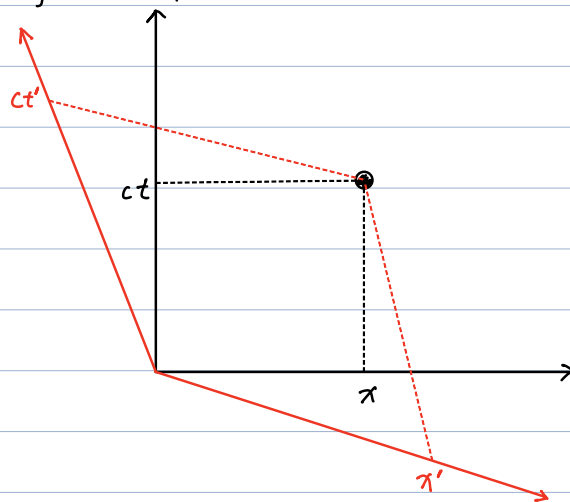
check: in the K frame, origin O' at time t:

$$\{ct, vt\}$$

$$\Rightarrow \begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct \\ vt \end{pmatrix} = \begin{pmatrix} \frac{ct}{\gamma} \\ 0 \end{pmatrix} \quad \checkmark$$



Q: what if train speed $-v$?



Velocity Addition :

$$w = u + v$$

↑ speed of ball relative to train
↓ speed of train

↑ speed of ball relative to track
↓

(Non-relativistic)

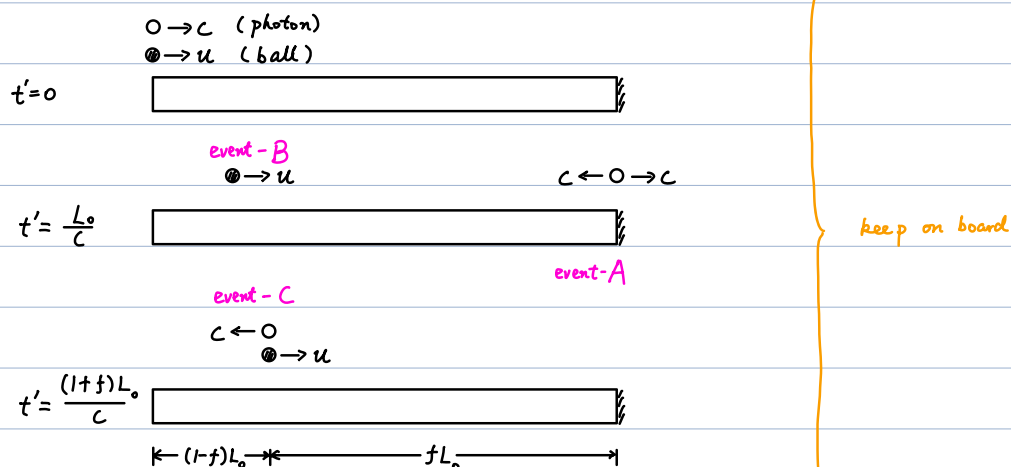
$$w = \frac{u + v}{1 + \frac{u}{c} \frac{v}{c}}$$

(Relativistic)

(Note: for $u, v \ll c \Rightarrow w = u + v$)

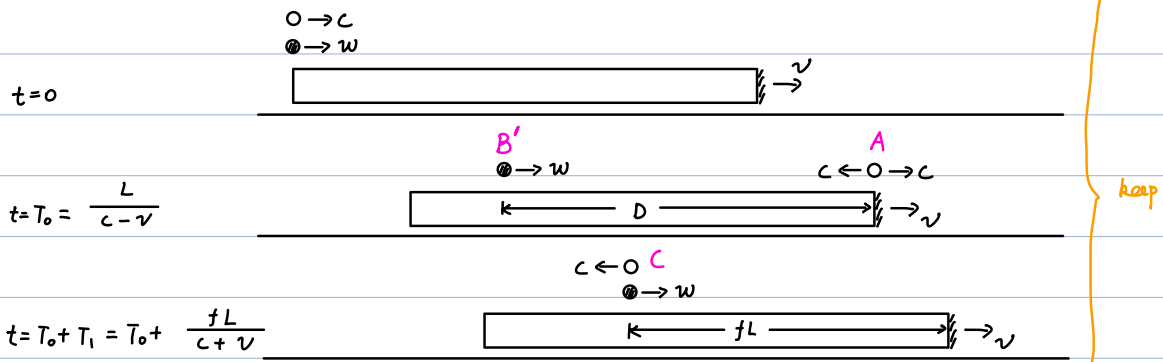
Proof (Method-1) :

① In the train frame (K')



$$\Rightarrow \frac{u}{c} = \frac{1-f}{1+f} \Rightarrow f = \frac{c-u}{c+u} \quad (*)$$

② In the track frame (K)



$$\begin{cases} D = cT_0 - wT_0 \\ D = cT_1 + wT_1 \end{cases}$$

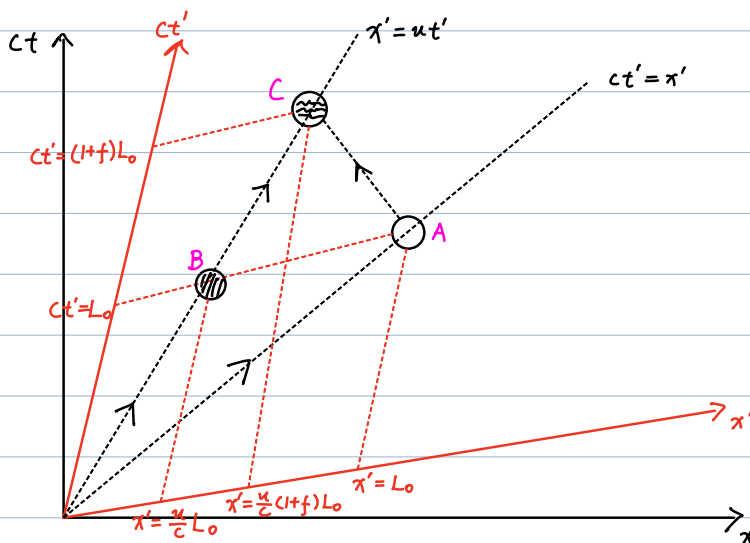
$$\Rightarrow \frac{T_0}{T_1} = \frac{c+w}{c-w}$$

$$\Rightarrow f \frac{c-v}{c+v} = \frac{c-w}{c+w} \quad (**)$$

Combine (*) (**) :

$$\Rightarrow \frac{c+v}{c-v} \frac{c-w}{c+w} = \frac{c-u}{c+u}$$

$$\Rightarrow w = \frac{u+v}{1 + \frac{u}{c} \frac{v}{c}} \quad \text{proved.}$$



Proof (Method-2) :

When photon & ball meets again,

$$\begin{cases} ct' = (1+f)L_0 \\ x' = \frac{u}{c}(1+f)L_0 \end{cases}$$

Lorentz transformation:

$$\begin{aligned} \begin{pmatrix} ct \\ x \end{pmatrix} &= \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} (1+f)L_0 \\ \frac{u}{c}(1+f)L_0 \end{pmatrix} \\ &= \gamma(1+f)L_0 \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \frac{u}{c} \end{pmatrix} \\ &= \gamma(1+f)L_0 \begin{pmatrix} 1 + \beta\frac{u}{c} \\ \beta + \frac{u}{c} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \Rightarrow w &= \frac{x}{t} \\ &= \frac{\beta + \frac{u}{c}}{(1 + \beta\frac{u}{c})/c} \\ &= \frac{u + v}{1 + \frac{uv}{c^2}} \quad \text{proved.} \end{aligned}$$

Proof (Method-3) :

$$\begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & & \\ \beta\gamma & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} c dt \\ dx \\ dy \\ dz \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & & \\ \beta\gamma & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} c dt' \\ dx' \\ dy' \\ dz' \end{pmatrix}$$

$$\Rightarrow \begin{cases} dt = \gamma dt' + \frac{\beta\gamma}{c} dx' \\ dx = \beta\gamma c dt' + \gamma dx' \\ dy = dy' \\ dz = dz' \end{cases}$$

$$\Rightarrow \frac{dx}{dt} = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} \quad \text{proved.}$$

Moreover,

$$\begin{cases} \frac{dy}{dt} = \frac{u'_y}{\gamma(1 + \frac{u'_x v}{c^2})} \\ \frac{dz}{dt} = \frac{u'_z}{\gamma(1 + \frac{u'_x v}{c^2})} \end{cases}$$

Comments :

- $u = c$

$$\Rightarrow w = \frac{c + v}{1 + \frac{v}{c}} = c \quad (\text{consistent with "constant speed of light"})$$

- $u = 0.9c, v = 0.9c$

$$\Rightarrow w = \frac{1.8c}{1 + 0.81} \approx 0.994c \quad (w \leq c)$$

(optional) Lorentz transformation if K' speed is $\vec{v} = v \hat{n}$ ($\hat{n}$ arbitrary direction)

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma n_x & -\beta\gamma n_y & -\beta\gamma n_z \\ -\beta\gamma n_x & 1 + (\gamma-1)n_x^2 & (\gamma-1)n_x n_y & (\gamma-1)n_x n_z \\ -\beta\gamma n_y & (\gamma-1)n_y n_x & 1 + (\gamma-1)n_y^2 & (\gamma-1)n_y n_z \\ -\beta\gamma n_z & (\gamma-1)n_z n_x & (\gamma-1)n_z n_y & 1 + (\gamma-1)n_z^2 \end{pmatrix}$$

Derivation :

$$R \hat{e}_x = \hat{n} \quad (\text{rotates } x\text{-axis to } \hat{n} \text{ direction})$$

define $\tilde{R} = \begin{pmatrix} 1 & \\ & R \end{pmatrix}$ (4x4 matrix)

Lorentz transformation matrix :

$$\Lambda = \tilde{R} \Lambda_x \tilde{R}^T$$

rotate back Lorentz along (new) x rotate to $\hat{n}$

$$\Lambda_x = \begin{pmatrix} \gamma & -\beta\gamma & & \\ -\beta\gamma & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$